

LATAM Bank Dataset

Factored Datathon 2026

Dataset Version 1.0.0

Total Records	~19 Million
Tables	13
Countries	Mexico, Colombia, Argentina
Date Range	June 2023 - June 2026

Generated: July 2026

Executive Summary

This dataset simulates a comprehensive **regional banking system** operating across three Latin American countries: Mexico, Colombia, and Argentina. It contains realistic banking, customer service, and digital interaction data spanning **three years** (June 2023 to June 2026).

The dataset includes **19,000,000 records** distributed across **13 tables**, covering everything from customer master data to transaction details, call center interactions, marketing campaigns, and customer complaints.

This is a **synthetically generated dataset** created specifically for the Factored Datathon 2026, with realistic Spanish names, addresses, and regional accent characteristics. The dataset intentionally includes data quality challenges (duplicates, nulls, late arrivals) to simulate real-world scenarios.

Key Characteristics

Attribute	Value
Total Records	19,000,000 rows
Total Tables	13
Countries	Mexico, Colombia, Argentina
Date Range	2023-06-17 to 2026-06-17
Currencies	MXN, COP, ARS, USD
Languages	Spanish (Mexican, Colombian, Argentine accents)

Data Quality Challenges

This dataset intentionally includes real-world data quality challenges to provide a realistic data engineering experience:

Challenge	Rate	Description
Duplicate Records	~2%	Realistic duplicate entries across tables
Null Values	~5%	Missing data in non-mandatory fields
Late Arrivals	Yes	Partitioned data may arrive late
Schema Evolution	Yes	Table schemas may evolve over time

Tables Overview

Dimension Tables

Table Name	Rows	Description
customers	150,000	Bank customer dimension table
products	400,000	Active financial products of customers
branches	350	Physical bank branches
service_agents	1,200	Customer service agents
marketing_campaigns	200	Bank marketing campaigns

Fact Tables

Table Name	Rows	Description
transactions	5,000,000	Daily financial transactions
call_center_interactions	800,000	Call center interactions with customers
call_transcripts	200,000	Call center call transcripts
satisfaction_surveys	250,000	Post-interaction satisfaction surveys (CSAT, NPS)
digital_events	10,000,000	Digital channel interaction events (mobile app, web)
complaints	80,000	Complaints and claims system (PQR)
campaign_sends	2,000,000	Individual marketing campaign sends

Reference Tables

Table Name	Rows	Description
daily_exchange_rates	3,000	Daily exchange rates for currency conversion

Potential Use Cases

Customer Analytics

- Customer segmentation and clustering
- Churn prediction models
- Customer lifetime value (CLV) analysis
- Cross-sell and up-sell opportunity identification

Contact Center Optimization

- First Call Resolution (FCR) improvement
- Agent performance analysis and benchmarking
- Sentiment trend analysis

- Accent-based routing optimization

Fraud Detection

- Transaction fraud detection models
- Anomaly detection in spending patterns
- Geographic risk modeling

Marketing Analytics

- Campaign effectiveness measurement
- Channel attribution modeling
- Personalization and targeting models
- A/B testing analysis

NLP / Text Analytics

- Topic modeling on call transcripts
- Intent classification
- Entity extraction from customer interactions
- Multilingual accent detection and classification

Product Analytics

- Product adoption and usage analysis
- Digital engagement funnel optimization
- Feature usage analysis

Important Notes

- **Spanish Language Data:** All text data (customer names, addresses, transcripts, descriptions) is in Spanish with regional variations.
- **Accent Detection:** The dataset includes accent detection fields (Mexican, Colombian, Argentine) enabling dialect-aware customer service analysis.
- **Multi-Currency Support:** All transactions include both local currency (MXN/COP/ARS) and USD conversion using daily exchange rates.
- **Date Partitioning:** Large fact tables are partitioned by date (year/month/day) for efficient processing and querying.
- **Referential Integrity:** Foreign key relationships are maintained across tables, though a small percentage of orphaned records may exist for testing.
- **Synthetic Data:** This is completely synthetic data generated for educational purposes. No real customer information is included.
- **Data Dictionary:** For detailed schema information including column descriptions, data types, and constraints, please refer to the complete DATA_DICTIONARY.md file.

For questions or support regarding this dataset,
please contact the Factored Datathon team.

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